LUCA VISINELLI



Tenure-track Assistant Professor (RTT)
Department of Physics "E.R. Caianiello"
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Scientific profile

Theoretical physicist working on particle cosmology and astroparticle physics, with a long-standing focus on QCD axions and light dark sectors. My research connects early-Universe particle production and dark matter structure to laboratory haloscopes, radio telescopes and gravitational probes. I have also developed research directions on primordial and astrophysical black holes, dark energy and gravitational waves (GWs).

Research indicators 96 journal articles published or accepted after peer review, plus 5 additional manuscripts under review; **h-index 50**; 15,000+ total citations; 28 papers with more than 100 citations and 50 with more than 50 citations (INSPIRE, October 27, 2026).

Awards NSFC Research Fund for International Excellent Young Scientists; Marie Skłodowska-Curie FELLINI Fellowship; 2021 Buchalter Cosmology Prize (Third Prize).

Current academic appointment

Since 2025 **Tenure-track Assistant Professor (Ricercatore a tempo determinato in tenure track, RTT)**, Department of Physics "E.R. Caianiello", Università degli Studi di Salerno, Italy; associated with INFN, Napoli g.c. Salerno.

Selected scientific contributions

QCD axion cosmology and nonlinear structure. Developed descriptions of axion dark-matter production in standard and non-standard cosmologies and studied the nonlinear evolution of miniclusters, streams and axion stars. Representative works: PRD 80 (2009); PRD 81 (2010); PLB 777 (2018); PRD 101 (2020); PRL 127 (2021).

Experimental searches of axions. Connected cosmological and astrophysical axion predictions to resonant haloscopes and radio-telescope searches; Theory WP Coordinator of FLASH, co-developer of ASTRA and initiator/co-proposer of the GBT Andromeda programme.

Black holes physics. Studied PBH–particle-DM coexistence, extremely massive primordial black holes, black-hole imaging, superradiance and Hawking-produced high-energy particles. Representative works: JCAP 2018; MNRAS 501, 506 (2021); PRD 100 (2019); PDU 49 (2025).

Light relics. Thermal/nonthermal axions, neutrino interactions, reionization and dark radiation.

Dark energy. Quintessence, cosmological tensions, scalar fifth forces and direct-detection probes.

Multi-messenger probes. Early-Universe GW production and high-frequency GW searches, including resonant cavities and quantum sensors.

Certifications

2025–2037 **Abilitazione Scientifica Nazionale (ASN)**, Professore di Prima Fascia, Settore 02/A2 Fisica Teorica delle Interazioni Fondamentali. [Decree accessible here]

Research leadership and group management

Since 2023 **Theory Work Package Coordinator, FLASH**, INFN Laboratori Nazionali di Frascati, Italy. Coordinates theoretical modelling and the scientific interpretation of axion and high-frequency gravitational wave searches.

Since 2024 **Co-developer, ASTRA**, University of Virginia, USA. Contributed to the development of the radio-telescope programme for axion dark matter searches, including source modelling, sensitivity forecasts, and observing strategies.

Since 2021 **Initiator and co-proposer, Green Bank Telescope axion searches**, USA. Initiated a programme searching for axion dark matter conversion in neutron-star magnetospheres and co-proposed the Andromeda observations GBT/22A-067 and GBT/23A-245.

2021–2025 **Research Group Leader**, Shanghai Jiao Tong University and Tsung-Dao Lee Institute, China. Established and led an independent research programme in astroparticle physics and cosmology, with emphasis on axion physics, dark matter, and the early Universe. Supervised three postdoctoral researchers and four Ph.D. students.

Education

December 16, 2011 **Ph.D. in Physics**, *The University of Utah*, Salt Lake City, USA.
Advisor: Prof. Paolo Gondolo. Thesis: *Axions in CDM and inflation models*.
Declarations and Ph.D. diploma available [at this link].
Italian recognition of the degree (*equipollenza al titolo di Dottore di ricerca*), Decree No. 0002642 of November 17, 2021 [link].

August 5, 2011 **M.Sc. in Physics**, *The University of Utah*, Salt Lake City, USA.
Advisor: Prof. Paolo Gondolo. GPA: 3.982/4.000.

June 22, 2007 **M.Sc. in Physics (Laurea specialistica)**, *University of Bologna*, Italy.
Advisor: Prof. Fiorenzo Bastianelli. Grade: 110/110 *cum laude*.
Declarations available [at this link].

October 14, 2005 **B.Sc. in Physics (Laurea triennale)**, *University of Bologna*, Italy.
Advisor: Prof. Giovanni Carlo Bonsignori. Grade: 110/110 *cum Laude*.
Declarations available [at this link].

July 4, 2002 **High School Diploma**, *High School "E. Fermi"*, Bologna, Italy. Grade: 100/100.

Experimental searches and observational programmes

2024–present **ASTRA** (Axion Search via Telescope for Radio Astronomy), University of Virginia, USA. ASTRA-low is a 0.4–2 GHz radio telescope at Fan Mountain Observatory, supported by the Jefferson Trust and approaching commissioning. ASTRA-high is a 2–20 GHz instrument funded by the U.S. National Science Foundation through MRI Award No. 2510185, led by Prof. Bradley Johnson. Co-author of the ASTRA sensitivity and science forecast [hep-ph/2603.13194].

2021–present **Green Bank Telescope axion searches**. Co-proposer of GBT/22A-067, an X-band search of Andromeda observed in 2022 and published in 2024, and GBT/23A-245, a C-band follow-up observed in 2023.

Since 2019 **FLASH** (FINUDA magnet for Light Axion SearcH), INFN Laboratori Nazionali di Frascati, Italy. Member of the Scientific Advisory Committee; the experiment pursues resonant searches for axions and high-frequency gravitational waves. Theory Work Package coordination is detailed above.

Previous academic appointments

2021–2025 **Tenure-Track Associate Professor**, Shanghai Jiao Tong University, Shanghai, China.
Joint Fellow, Tsung-Dao Lee Institute, Shanghai, China.

2020–2021 **Marie Skłodowska-Curie FELLINI Fellow**, INFN Laboratori Nazionali di Frascati, Italy.
Host principal investigator: Dr. Enrico Nardi.
Contract and declarations available [at this link].

2019–2020 **GRAPPA Fellow**, GRAPPA, University of Amsterdam, Netherlands.
Principal investigator: Prof. Christoph Weniger.
Contract and declarations available [at this link].

2016–2019 **Postdoctoral Researcher**, Nordita and Stockholm University, Sweden.
Principal investigator: Prof. Katherine Freese.
Contract and declarations available [at this link].

2013–2015 **Postdoctoral Fellow**, Euro-Mediterranean Center on Climate Change (CMCC), Bologna IT.
Principal investigator: Prof. Simona Masina.
Contract and declarations available [at this link].

2007–2011 **Research Assistant**, The University of Utah, USA.

Individual grants and fellowships

2023–2025 **Research Fund for International Excellent Young Scientists**, National Natural Science Foundation of China (NSFC), Grant No. 12350610240, "Astrophysical Axion Laboratories".
Principal Investigator: **Luca Visinelli**. Awarded August 2023.
Funding: approximately EUR 110,000 over two years.
Grant contract accessible [at this link].

2021–2027 **Start-up funding**, Tsung-Dao Lee Institute, China; award period stated in the November 2020 offer.
Principal Investigator: **Luca Visinelli**.
Funding offered: approximately EUR 210,000.
Offer letter accessible [at this link].

2020–2023 **Horizon 2020 FELLINI programme**, Marie Skłodowska-Curie COFUND Action, Grant Agreement No. 754496 (H2020-MSCA-COFUND-2016 FELLINI), "Tools for Axions, Leptogenesis, and Neutrino Theories (TALeNT)".
Principal Investigator: **Luca Visinelli**. Awarded November 2020; project period as stated in the award.
Funding: EUR 50,520 per year.
Grant contract accessible [at this link].

2025–2026 **State Key Laboratory of Dark Matter Physics**, Tsung-Dao Lee Institute, China. Awarded approximately EUR 7,000 in personal research funding.

2007–2008 **Award for Ph.D. students abroad**, University of Bologna, Italy.
Funding: EUR 26,318.70.
Grant contract accessible [at this link].

Fall 2006 **Undergraduate Student Award**, University of Bologna, Italy. Funding: approximately EUR 2,000.

Awarded observing time

2022 **Green Bank Telescope Proposal GBT/23A-245**, "Continuing the Search for Axion Dark Matter in Andromeda with VEGAS". Accepted; observations completed in 2023.

August 2021 **Green Bank Telescope Proposal GBT/22A-067**, "Searching for Axion Dark Matter in Andromeda Using an X-band Receiver". Accepted; observations completed in 2022.

Participation in collaborative research projects

2025–2026 **SCOPE – Screened Couplings and Pauli Exclusion: Probing Fundamental Physics with Light Fields**, external research project proposal to the State Key Laboratory of Dark Matter Physics. Project leader: Prof. Antonino Marcianò (Fudan University). Research on environmentally screened light scalar fields, stellar production mechanisms, searches for violations of the Pauli Exclusion Principle.
Requested funding: CNY 100,000; proposed duration: July 2025–December 2026.

2025–present **INFN research activity allocation**: QGSKY (Iniziativa Specifica; 60% FTE), QUAX (20% FTE), and Einstein Telescope (ET; 20% FTE).

2019–2020 **Dutch Research Council**, Contract No. 680.92.18.03, "The Hidden Universe of Weakly Interacting Particles" (2018–2023).
Project Principal Investigator: Prof. Paul De Jong, University of Amsterdam.
Role: GRAPPA Fellow, University of Amsterdam.

2016–2019 **Swedish Research Council**, Contract No. 638-2013-8993 (2014–2024).
Project Principal Investigator: Prof. Katherine Freese. Total project funding: approximately EUR 10,000,000.
Role: Postdoctoral researcher at Stockholm University and Uppsala University.

2014–2015 **EU FP7-SPACE, "MyOcean follow-up"**, Euro-Mediterranean Center on Climate Change (CMCC), Bologna, Italy. Project Principal Investigator: Prof. Simona Masina. Total project budget: EUR 99,244.19. Role: Postdoctoral researcher at CMCC.

2012–2014 **EU FP7-SPACE, "MyOcean2"**, CMCC, Bologna, Italy.
Project Principal Investigator: Prof. Simona Masina. Total project budget: EUR 464,980. Participation during the CMCC appointment (2013–2015).

2012–2014 **EU FP7-ENV, "GEOCARBON"**, CMCC, Bologna, Italy.
Project Principal Investigator: Prof. Simona Masina. Total project budget: EUR 95,000. Participation during the CMCC appointment (2013–2015).

Prizes

January 2022 2021 Buchalter Cosmology Prize (Third Prize) for the paper "The XENON1T excess and future prospects" [hep-ph/2103.15834]. Also announced by the American Astronomical Society. Co-recipients: S. Vagnozzi, P. Brax, A.C. Davis, J. Sakstein.

Additional distinctions

Spring 2025 Italian tax-relief scheme for returning researchers ("Rientro dei Cervelli").

Fall 2019 Dutch tax-relief scheme for highly skilled migrants (30% ruling).

Fall 2016 Swedish tax relief for foreign key personnel.

Spring 2011 **Outstanding Teaching Assistantship Award**, The University of Utah.

2001–2002 **Honorable Mention**, Italian Physics Olympiads.

Participation in COST Actions and proposals

2026 COST Action proposal OC-2025-1-28663 "Dark Waves", submitted.
Principal Investigator: Prof. Alessandro Mirizzi.
Contributed to the original concept of the proposal; invitation available [at this link].

2026 COST Action proposal OC-2025-1-28795 "NEXTCOSMO", submitted.
Principal Investigator: Dr. Eleonora Di Valentino.
Contributed to the original concept of the proposal; invitation available [at this link].

2022 COST Action CA21136 "CosmoVerse", approved.
Principal Investigator: Dr. Eleonora Di Valentino.

2022 COST Action CA21106 "COSMIC WISPers in the Dark Universe", approved.
Principal Investigator: Prof. Alessandro Mirizzi.
Contributed to the original concept of the Action; invitation available [at this link].

Academic Service

2025 **Ph.D. Dissertation Committee Member**. Served on dissertation committees in New Zealand and Spain.

2025 **MSCA Fellowship Evaluation**. Invited evaluator for proposals under the Marie Skłodowska-Curie Actions programme.

2025 **Master's Degree Committee Member**, Department of Physics, Università degli Studi di Salerno. Served on the Physics graduation committee on October 22, 2025.

2021–2025 **Committee member**, Tsung-Dao Lee Institute and Shanghai Jiao Tong University. Regularly served on committees selecting graduate students, postdoctoral researchers, junior faculty, and recipients of student and research awards.

2024–2025 **Member of CSC** (China Scholarship Council), Shanghai Jiao Tong University. Assessed students for international mobility funding.

2024 **External committee member** for MSc degrees, Fudan University (Shanghai).

2023 **Co-organizer** of the "Tea time" colloquia at the Tsung-Dao Lee Institute.

2017–2018 **Postdoctoral Representative**, Nordita, Stockholm (Sweden).

2017–2018 **Co-organizer** of the "Beyond the Standard Model" workgroup at Stockholm University.

2016–present Regular referee for Physical Review Letters (PRL), Physical Review D (PRD), Physics Letters B (PLB), Physics of the Dark Universe (PDU), Journal of Cosmology and Astroparticle Physics (JCAP), Modern Physics Letters A (MPLA), and Universe.
Number of reports per year from www.webofscience.com available [at this link]

Editorial service

2026–present **Editor**, Physics of the Dark Universe, Elsevier.

2024–2025 **Symmetry**, Section Board Member: editorial-board profile.

2020–2022 Guest editor for the Special Issue of the journal Universe: *"Dark Matter and Dark Energy: Particle Physics, Cosmology, & Experimental Searches"* (2024).
Link: www.mdpi-com/journal/universe/special_issues/DM_DE.

Organization of international conferences and workshops

2027 (forthcoming) **Organizational involvement**, SIGRAV International School 2027, Vietri sul Mare, Italy.

February 2027 (forthcoming) **Chairperson**, GPQ-Bridges: Bridging Gravity to Quantum Mechanics through Probability, ECT*, Trento, Italy, February 1–5, 2027.

June 2026 **Co-organizer**, Exploring New Frontiers in Cosmology: The Quest for a More Comprehensive Concordance Model, Galileo Galilei Institute, Arcetri, Italy.

March 2026 **Co-organizer**, PAFT2026, Vietri sul Mare, Italy.

November 2025 **Convener**, TeVPA 2025, Valencia, Spain.

November 2024 **Scientific Organizing Committee Member**, Shanghai Symposium on Particle Physics and Cosmology (SPCS 2024), Tsung-Dao Lee Institute, Shanghai, China. Supporting documentation available [here].

December 2023 **Local Organizing Committee Member**, 32nd Texas Symposium on Relativistic Astrophysics, Tsung-Dao Lee Institute, Shanghai, China. Responsible for organizing the Dark Matter mini-symposium. Supporting documentation available [here].

October 2023 **Organizer**, Advancements in Axion Physics, Tsung-Dao Lee Institute, Shanghai, China. Recorded presentations available [here].

Technical skills and scientific computing

Code Python, Fortran, C/C++, R, Visual Basic, and Pascal; parallel computing with MPI.

Scientific tools Mathematica, MATLAB, GALPROP, DarkSUSY, Gadget-2, MESA, NEMO, and BFM.

Data and workflows Large scientific datasets in NetCDF and ASCII formats; NCO and CDO; version control and collaborative code maintenance with Git.

HPC Linux and UNIX environments; experience with distributed-memory systems, including the 7,712-core IBM iDataPlex "Athena" cluster.

Scientific writing LaTeX, Microsoft Office, and Vi; working familiarity with Windows and macOS.

Teaching at the University of Salerno

Spring 2027 Graduate course "Cosmology and Astroparticle Physics" (10 hours)

Spring 2027 Fisica ed elementi di matematica, Corso di Laurea in Farmacia (6 CFU)

Fall 2026 Fisica con elementi di matematica e statistica, Corso di Laurea in Farmaceutica (3 CFU)

Spring 2026 Graduate course "Cosmology and astroparticles" (10 hours)

Spring 2026 Laboratorio di Fisica II, Corso di Laurea in Matematica (4 CFU)

Spring 2026 Laboratorio di Fisica I, Corso di Laurea in Chimica (4 CFU)

Spring 2025 Fisica I, Corso di Laurea in Chimica (2 CFU)

Spring 2025 Laboratorio di Meccanica e Termodinamica, Corso di Laurea in Fisica (1 CFU)

Additional teaching experience

Fall 2017 Lecturer for FK5024 "Nuclear physics", Stockholm University, Stockholm (Sweden).
The course reflections are available [at this link].

2015–2016 Lecturer for "Introductory Mathematics", Department of Political Sciences, University of Bologna (Italy)
The documentation proving the enrollment is available [at this link].

2015 Teaching assistant in Mathematics for International Markets, Department of Economics, University of Bologna. Supervisor: Prof. Sabrina Mulinacci
The documentation proving the enrollment is available [at this link].
The contribution is acknowledged [on the course webpage].

2015 Teaching assistant for "Mathematics for Economics and Finance", Department of Economics, University of Bologna. Supervisor: Prof. Alessandra Giovagnoli
The documentation proving the enrollment is available [at this link].

2010–2011 Teaching assistant for Physics 3740, "Special relativity and quantum mechanics", The University of Utah. Supervisors: Prof. Kyle Dawson and Prof. Jordan Gerton

2009 Teaching assistant for Physics 5020, "Electromagnetism", The University of Utah. Supervisor: Prof. Mikhail Raikh

2008 Teaching assistant for Physics 5010, "Classical and Quantum Mechanics", The University of Utah. Supervisor: Prof. Mikhail Raikh

Supervision of postdoctoral researchers

2025–present Md Riajul Haque, Tsung-Dao Lee Institute. INSPIRE profile.

2024–2026 Hong-Yi Zhang, Tsung-Dao Lee Institute. https://inspirehep.net/1790638.
Now faculty.

2022–2024 Michael Zantedeschi, Tsung-Dao Lee Institute. https://inspirehep.net/1859878.
Now researcher at INFN Pisa, Italy.

Supervision and mentoring of Ph.D. students

2025–present Miriam Bulliri, University of Salerno & INFN.

2023–present Hanyu Cheng, Tsung-Dao Lee Institute and Shanghai Jiao Tong University.

2023–present Ziwen Yin, Tsung-Dao Lee Institute and Shanghai Jiao Tong University.

2024–2025 Xinhui Chu, Tsung-Dao Lee Institute & Shanghai Jiao Tong University

2024–2025 Yongzhi Tang, Tsung-Dao Lee Institute & Shanghai Jiao Tong University

Supervision and mentoring of undergraduate and master's students

2023–2024 Qixuan Xu, leap year at Tsung-Dao Lee Institute (China). He is now a Ph.D. candidate at Instituto Superior Técnico, Lisbon.

2022–2023 Kratika Mazde, M.Sc., Indian Institute of Science Education and Research. She is now a doctoral researcher at the Institut d'Astrophysique de Paris. Her thesis led to a publication in JCAP.
Kratika's M.Sc. thesis with my mention is found [at this link].

2019–2020 Nicklas Ramberg, M.Sc., Uppsala University. Now a postdoctoral researcher at SISSA. M.Sc. thesis acknowledging my supervision available [at this link].

Informal Mentoring of PhD and M.Sc. students

2023–2024 Andrea Boccia (Ph.D. student, University of Napoli). Main advisor: Fabio Iocco.

2021–2023 Rittick Roy (M.Sc. student, Fudan University). Main advisor: Cosimo Bambi.

2019–2022 Youjia Wu (Ph.D. student, University of Michigan). Main advisor: Katherine Freese.

2017–2018 Janina Renk (Ph.D. student, Stockholm University). Main advisor: Joakim Edsjö.

Academic mentoring and reference support

Andrea Boccia, Luca Caloni, Hanyu Cheng, Kratika Mazde, Nicklas Ramberg, Salvatore Samuele Sirletti, Rittick Roy, Sunny Vagnozzi, Qixuan Xu, Ziwen Yin, Michael Zantedeschi, Hong-Yi Zhang

Bibliometric summary and research identifiers

96 journal articles published or accepted after peer review, plus 5 additional manuscripts under review; **h-index 50**; 15,000+ total citations; 21 papers with more than 100 citations and 43 with more than 50 citations (INSPIRE statistics as of October 27, 2026). Publication status is reported explicitly in the complete lists below. See also INSPIRE, ADS, and Google Scholar.

ORCID 0000-0001-7958-8940

WoS ID J-5573-2015

Scopus 34168444500

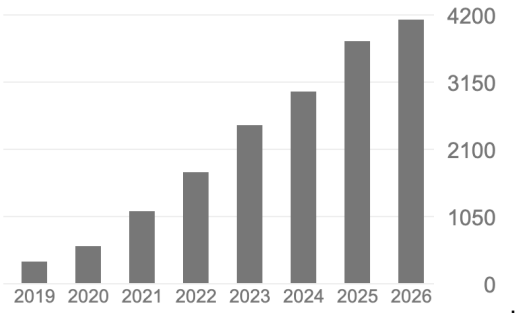
arXiv Complete author listing

SciProfiles Author profile

GitHub github.com/lucavisinelli

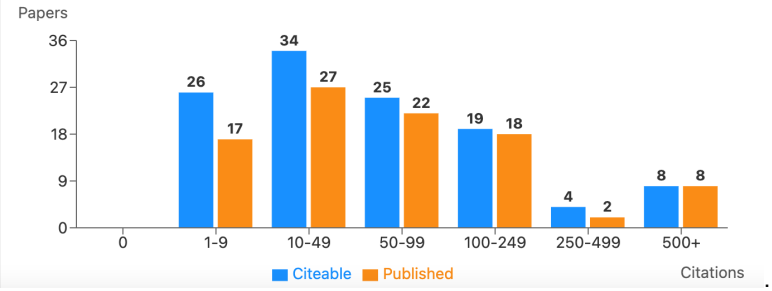
Citation statistics as of October 27, 2026

	All	Since 2021
Citations	17826	16359
h-index	54	49
i10-index	93	88



Citations per year (Google Scholar)

	Citeable	Published
Papers	**116**	**94**
Citations	**16,195**	**15,077**
h-index	**52**	**50**
Citations/paper (avg)	**139.6**	**160.4**



Total number of citations (HEP-INSPIRE)

Articles published in peer-reviewed international journals

A complete list of my publications can be found at: http://inspirehep.net/author/profile/L.Visinelli.1

96. D. Amaral *et al* (with **L. Visinelli**),
Global detector network to search for high-frequency gravitational waves (GravNet): conceptual design,
Phys. Rev. D [astro-ph/2603.24645] (2026).

95. Z. Yin, H. Cheng, and **L. Visinelli**,
Primordial Black Hole Abundance from Reionization,
Phys. Dark Univ. **54**, 102487 [astro-ph/2602.06794] (2026).

94. Md Riajul Haque, Fabio Iocco, and **L. Visinelli**,
Primordial Black Hole interpretation of the sub-solar merger event S251112cm,
Phys. Lett. B **882**, 141049 [astro-ph/2603.25795] (2026).

93. **L. Visinelli** and M. Naydenov,
Transient axion streams from disrupted miniclusters,
Phys. Rev. D **114**, 055056 [astro-ph/2605.28005] (2026).

92. H. Banks, A.-C. Davis, and **L. Visinelli**,
Probing Solar Symmetrons with Direct Detection,
Phys. Rev. D **114**, 055029 [hep-ph/2604.09283] (2026).

91. Z. Yin, H. Cheng, E. Di Valentino, N. Gendler, D. J. E. Marsh, and **L. Visinelli**,
Constraining the axiverse with reionization,
Phys. Rev. D [hep-ph/2507.03535] (2026).

90. G.F. Vitale, G. Lambiase, T.K. Poddar, and **L. Visinelli**,
Microscopic primordial black holes as macroscopic dark matter from large extra dimensions,
JCAP **2608**, 025 [astro-ph/2604.14871] (2026).

89. N. Barbieri, L. Caloni, M. Gerbino, M. Lattanzi, and **L. Visinelli**,
Beyond thermal approximations: Precise cosmological bounds on Axion-Like Particles,
JCAP **2607**, 099 [astro-ph/2602.11100] (2026).

88. Y.-F. Cai, **L. Visinelli**, and S.-F. Yang,
Atomic Quantum Sensors for High-Frequency Gravitational Wave Searches,
Phys. Dark Univ. **53**, 102374 [astro-ph/2510.15031] (2026).

87. G.-W. Yuan, A.-C. Davis, M. Giannotti, S. Vagnozzi, **L. Visinelli**, and J. Vogel,
Direct detection of solar chameleons with electron recoil data from XENONnT,
Phys. Rev. D **113**, 123024 [hep-ph/2511.01655] (2026).

86. H. Cheng, E. Di Valentino, and **L. Visinelli**,
Cosmic Strings as Dynamical Dark Energy: Novel Constraints,
J. High Energy Astrophys. **53**, 100610 [astro-ph/2505.22066] (2026).

85. M. Figliolia, F. Grippa, G. Lambiase, and **L. Visinelli**,
Gravitational Signatures of Axion Dark Matter via Parity-Violating Interactions,
Phys. Rev. D **113**, 023012 [astro-ph/2509.12038] (2026).

84. W. Lin, **L. Visinelli**, and T.T Yanagida,
Testing Quintessence Axion Dark Energy with Recent Cosmological Results,
JCAP **2510**, 023 [astro-ph/2504.17638] (2025).

83. H.-Y. Zhang, P. Arias, A. Cheek, E. Schiappacasse, **L. Visinelli**, and L. Roszkowski,
Dark photon dark matter from flattened axion potentials,
J. High Energ. Phys. **2025**, 142 [hep-ph/2507.20484] (2025).

82. H. Cheng, E. Di Valentino, L. Escamilla, A. Sen, and **L. Visinelli**,
Pressure Parametrization of Dark Energy: First and Second-Order Constraints with Latest Cosmological Data, JCAP **2509**, 031 [astro-ph/2505.02932] (2025).

81. A. Cheek, **L. Visinelli**, and H.-Y. Zhang,
Testing the dark origin of neutrino masses with oscillation experiments,
Phys. Rev. Lett. **135**, 031801 [hep-ph/2503.08439] (2025).

80. M. Zantedeschi and **L. Visinelli**,
Ultralight Black Holes as Sources of High-Energy Particles,
Phys. Dark Univ. **49**, 102034 [astro-ph/2410.07037] (2025).

79. G. Lambiase, T.K. Poddar, and **L. Visinelli**,
Impact of the cosmic neutrino background on black hole superradiance,
Phys. Rev. D **112**, 016010 [hep-ph/2503.02940] (2025).

78. D. Ayzenberg *et al* (with **L. Visinelli**),
Fundamental physics opportunities with future ground-based mm/sub-mm VLBI arrays,
Living Rev. Relativ. **28**, 4 [astro-ph/2312.02130] (2025).

77. E. Di Valentino *et al,* (**L. Visinelli** *as Section Coordinator*),
The CosmoVerse White Paper,
Phys. Dark Univ. **49**, 101965 [astro-ph/2504.01669] (2025).

76. A. Boccia, F. Iocco, and **L. Visinelli**,
Constraining the primordial black hole abundance through Big-Bang nucleosynthesis,
Phys. Rev. D **111** 063508 [astro-ph/2405.18493] (2025).

75. **L. Visinelli**,
Inflating without a flat potential: Viscous inflation,
Phys. Dark Univ. **48**, 101921 [gr-qc/1604.03873] (2025).

74. L. Walters *et al* (with **L. Visinelli**),
Axions in Andromeda: Searching for minicluster-neutron star encounters with the Green Bank Telescope,
Phys. Rev. D **110**, 123002 [astro-ph/2407.13060] (2024).

73. Y.-D. Tsai, D. Farnocchia, M. Micheli, S. Vagnozzi, and **L. Visinelli**,
Constraints on fifth forces and ultralight dark matter from OSIRIS-REx target asteroid Bennu,
Nature Commun. Phys. **7**, 311 [hep-ph/2309.13106] (2024).

72. **L. Visinelli**, T.T. Yanagida, and M. Zantedeschi,
Do neutrinos bend? Consequences of an ultralight gauge field as dark matter,
Phys. Dark Univ. **46**, 101659 [hep-ph/2407.18300] (2024).

71. Z. Yin and **L. Visinelli**,
Axion star condensation around primordial black holes and microlensing limits,
JCAP **2410**, 013 [gr-qc/2404.10340] (2024).

70. H. Cheng and **L. Visinelli**,
Future targets for light gauge bosons from cosmic strings,
Phys. Dark Univ. **46**, 101667 [hep-ph/2408.16334] (2024).

69. T. O'Shea, A.-C. Davis, M. Giannotti, S. Vagnozzi, **L. Visinelli**, and J. Vogel,
Solar chameleons: novel channels (I),
Phys. Rev. D **110**, 063027 [hep-ph/2406.01691] (2024).

68. C. Gatti, **L. Visinelli**, M. Zantedeschi,
Cavity Detection of Gravitational Waves: Where Do We Stand?,
Phys. Rev. D **110**, 023018 [gr-qc/2403.18610] (2024).

67. F. Iocco and **L. Visinelli**,
Compatibility of JWST results with exotic halos,
Phys. Dark Univ. **44**, 101496 [astro-ph/2403.13068] (2024).

66. S. Bulanov *et al* (with **L. Visinelli**),
The science case for an intermediate energy advanced and novel accelerator linear collider facility,
JINST **19**, T01010 [physics.acc-ph/2203.08425] (2024).

65. G. Montefalcone, V Aragam, **L. Visinelli**, and K. Freese,
`WarmSPy`**:** *A numerical study of cosmological perturbations in warm inflation*,
JCAP **2401**, 032 [astro-ph/2306.16190] (2024).

64. A. Addazi, Y.-F. Cai, A. Marcianò, and **L. Visinelli**,
Have pulsar timing array methods detected a cosmological phase transition?,
Phys. Rev. D **109**, 015028 [astro-ph/2306.17205] (2024).

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Invited articles

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Preprints under review

5. **L. Visinelli**,
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4. U. Bhura *et al* (with **L. Visinelli**),
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2. A. Mirizzi et al. (**L. Visinelli** *as Contributor*),
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1. H. Cheng, Z. Yin, E. Di Valentino, D. J. E. Marsh, and **L. Visinelli**,
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5. **L. Visinelli**,
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4. **L. Visinelli** *et al*,
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Articles in preparation

3. G.F. Vitale, G. Lambiase, T.K. Poddar, and **L. Visinelli**,
Microscopic primordial black holes as macroscopic dark matter from large extra dimensions II, in preparation.

2. M. Bulliri and **L. Visinelli**, *2HDM*, in preparation.

1. M. Bulliri and **L. Visinelli**, *Dark Radiation from a Dynamical Radial Mode*, in preparation.

Technical reports

5. *Endorser*, European Strategy of Particle Physics,
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4. R. X. Adhikari *et al*, *Report of the Topical Group on Cosmic Probes of Fundamental Physics for Snowmass 2021*
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3. J. Jaeckel, G. Rybka, L. Winslow, *et al*, *Axion Dark Matter*
[hep-ex/2203.14923] (2022).

2. D. Antypas *et al*, *New Horizons: Scalar and Vector Ultralight Dark Matter*
[hep-ex/2203.14915] (2022).

1. D. Alesini *et al*, *KLASH Conceptual Design Report*
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Invited talks

69. 11/02/27 Quantum Seminars, Mainz Institute for Theoretical Physics. **TBD**.
68. 01/09/26 University of Science and Technology of China (USTC), Hefei (CHN). Searching for Axion Dark Matter in the Laboratory and the Sky.
67. 02/07/26 Technische Universität München (DE), HEP group meeting. New opportunities in axion cosmology.
66. 25/03/26 DESY Hamburg (DE), MADMAX Collaboration Meeting. Axion cosmological scenarios, miniclusters, and their tidal disruption.
65. 30/09/25 University of Sofia (BG), Cosmological Tensions: Overview of Theory Solutions.
64. 09/09/25 Tsung-Dao Lee Institute (CHN), New opportunities in axion cosmology.
63. 03/12/25 Università di Napoli, Napoli (IT), Searching for Axion Dark Matter with Radio Telescopes.
62. 02/28/25 INFN Laboratori Nazionali di Frascati, Frascati (IT), Indirect detection of the QCD axion.
61. 01/16/25 INFN Università di Pisa, Pisa (IT), Indirect and direct detection of the QCD axion.
60. 12/30/24 ShanghaiTech University, Shanghai (CHN), Theoretical motivations and phenomenology of light bosons.
59. 10/25/24 Fudan University, Shanghai (CHN), The interplay between primordial BHs and particle DM.
58. 09/13/24 University of Torino (IT), Indirect Detection of the QCD Axion.
57. 06/04/24 University of Trento (IT), Theoretical motivations for a light boson and its phenomenology.
56. 05/27/24 Institut D'Astrophysique de Paris (FR), The interplay between primordial BHs and particle DM.
55. 03/28/24 Yunnan University (CHN), Axion Resonant Conversion in Neutron Star Magnetospheres.
54. 03/13/24 IBS Science and Culture Center, Daejeon (Korea), Indirect Detection of the QCD Axion: Resonant Conversion in Neutron Star Magnetospheres
53. 03/06/24 Shanghai Jiao Tong University (CHN), Indirect Detection of the QCD Axion: Resonant Conversion in Neutron Star Magnetospheres
52. 01/31/24 University of Louisiana at Lafayette (USA), Tracing the footprints of dark matter
51. 08/22/23 University of Texas at Austin (USA), Axions as dark matter or dark radiation
50. 07/18/23 University of Bologna (IT), The interplay of primordial black holes, dark matter, and axions
49. 05/16/23 Fudan University, Shanghai (CHN), Dark radiation in the Universe
48. 03/09/23 University of Napoli (IT), The interplay of primordial black holes, dark matter, and axions
47. 02/09/23 University of Trento (IT), The interplay of primordial black holes and particle dark matter
46. 02/06/23 University of Ferrara (IT), The interplay of primordial black holes and particle dark matter
45. 01/26/23 DESY (DE), Novel Cosmological Bounds on Thermally-produced Axion-like Particles

44. 05/20/22 Chinese Academy of Science, Beijing (CHN), Radio signals from axions conversion
43. 04/27/22 KEK, Tsukuba (JP), Radio signals from axions conversion
42. 04/08/22 Indian Institute of Technology, Mumbai (IN), Superradiance evolution of black hole shadows
41. 12/15/21 Helsinki Institute of Physics (FI), Axion miniclusters in the Milky Way
40. 10/14/21 SISSA, Trieste (IT), Radio signals from axion miniclusters colliding with neutron stars
39. 06/11/21 University of L'Aquila (IT), The population of axion miniclusters in the Galaxy
38. 04/26/21 Shanghai Jiao Tong University, Shanghai (CHN), Direct detection of dark energy
37. 02/17/21 Kavli IPMU, Tokyo (JP), *The interplay of primordial black holes and particle dark matter*
36. 11/04/20 Institute of Physics, Academia Sinica (TW), *Compact objects and dark matter*
35. 10/13/20 Undergraduate seminar, Colgate University (USA), *One dark matter candidate: the axion*
34. 06/04/20 International Institute of Physics, Natal (BR), *Radio and gravitational wave signals from cosmic axions* [*LINK to the YouTube video on the IIP channel: https://rb.gy/tsg0of*]
33. 05/11/20 Newton 1665 webinars series, *New physics out of the Shadow*
[*LINK to the YouTube video on the Newton1665 channel: https://www.youtube.com/watch?v=yCDUfzv8oKY*]
32. 04/29/20 Latin American Webinars (LAWphysics) series, *Astrophysics with axion stars and miniclusters*
[*LINK to the YouTube video on the LAWphysics channel: https://www.youtube.com/watch?v=ilfmBKMgyH8*]
31. 04/07/20 University of Texas at Austin (USA), *The future of light boson dark matter*
30. 03/02/20 DAMTP Institute of Astronomy, University of Cambridge (UK), *Light boson dark matter*
29. 12/20/19 National Institute Of Chemical Physics And Biophysics, Tallinn (EE), *Light boson dark matter*
28. 11/27/19 Fudan University, Shanghai (CHN), *Testing the rotational nature of the supermassive object M87**
27. 11/25/19 Shanghai Jiao Tong University, Shanghai (CHN), *Light bosons as dark matter candidates*
26. 11/07/19 Nordita, Stockholm (SE), *Testing the rotational nature of the supermassive object M87**
25. 05/21/19 INFN Frascati, Rome (IT), *Axion miniclusters and implications for axion detection*
24. 05/10/19 Wayne State University, MI (USA), *Probing the Early Universe with Axion Physics*
23. 05/01/19 Kavli Institute for Cosmological Physics, IL (USA), *Probing the Early Universe with Axions*
22. 04/30/19 Argonne National Laboratory, IL (USA), *Probing the Early Universe with Axions*
21. 04/25/19 University of Michigan, MI (USA), *The Quest for the Axion*
20. 04/17/19 Barry University, FL (USA), *Introduction to Cosmology and Particle Physics*
19. 04/12/19 University of Florida, FL (USA), *Probing the Early Universe with Axion Physics*
18. 04/04/19 IFIC, Valencia (ES), *Probing the Early Universe with Axions*
17. 01/08/19 SISSA, Trieste (IT), *The Cold Dark Matter axion and Axion Stars*
16. 10/23/18 Nikhef Amsterdam (NL), *The Quest for the Axion*

15. 11/30/18 INFN Frascati, Rome (IT), *Motivations for the search of light axions*
14. 10/23/18 University of Bologna (IT), *Searching for Axions and the String Axiverse in the Cosmo*
13. 10/18/18 INFN Frascati, Rome (IT), *Searching for Axions in the Lab and in the Cosmo*
12. 04/11/18 Latin American Webinars (LAWphysics) series, *The axion in cosmology and astrophysics* [*LINK to the YouTube video on the LAWphysics channel: https://www.youtube.com/watch?v=YWqVpPrpLjw*]
11. 02/01/18 CEICO, Prague (CZ), *The Cold Dark Matter axion and Axion Stars*
10. 12/01/17 University of Torino (IT), *The Cold Dark Matter axion and Axion Stars*
9. 08/16/17 University of Oslo (NO), *Dark matter capture and neutrino observatories*
8. 06/15/17 University of Bologna (IT), *Dark matter capture and neutrino observatories*
7. 02/01/17 University of Michigan, MI (USA), *Axion cold dark matter, miniclusters, and axion stars*
6. 10/25/16 University of Helsinki (FI), Axion cosmology.
5. 01/21/16 University of Zaragoza (ES), *Axion cold dark matter, status and perspectives.*
4. 06/03/13 CMCC Lecce (IT), *Impacts of Data Assimilation on the Global Ocean Carbonate System*
3. 06/15/11 University of Pisa (IT), *Axion cold dark matter in standard and non-standard cosmologies*
2. 10/12/10 University of New Mexico, NM (USA), *An integral equation for distorted-wave amplitudes*
1. 05/21/09 University of Bologna (IT), *Axion cold dark matter revisited*

Conference talks

62. 18–22/01/27: Mechanical Quantum Sensors for New Physics Searches, Mainz Institute for Theoretical Physics (DE), *TBD*
61. 07/09 26: New directions EXploring the Unknown beyond the Standard Model (NEXUS 2026), Shanghai (CHN), *Implications of a μeV axion discovery*
60. 01/07 26: FLASH Founding Meeting, Laboratori Nazionali di Frascati (Roma), *Cosmological axion at the micro-eV*
59. 06/8-12 26: Beyond WIMPs: Theory and model building for the next generation of dark matter searches, Pollica (SA) (Italy), *Implications of a μeV axion discovery*
58. 25/02 26: FLASH Meeting, Laboratori Nazionali di Frascati (Roma), *Axion streams are not bound to stay*
57. 25/11 25: FLASH Meeting, Laboratori Nazionali di Frascati (Roma), *HFGW detection in resonant cavities*
56. 11/13-14 25: Annual QGSKY Meeting, Catania (Italy), *New avenues to detect light bosons and gravitational waves*
55. 11/3-7 25: TeV Particle Astrophysics (TeVPA) 2025, Valencia (Spain), *Searching for Axion Dark Matter with Radio Telescopes*

54. 10/27-29 25: **Invited speaker**, National Space Forum 2025, Sofia (Bulgaria), *Constraints on Exotic Early-Universe Physics from the CMB*

53. 09/1-7 25: FLASH Meeting, Laboratori Nazionali di Frascati (Roma), *Cosmic Dark Photon and Axion Radiation and Streams*

52. 09/1-7 25: Particle Physics on Tabletops (PPTT), Chengdu (CHN), *Tabletop Tests of Modified Gravity: Constraints and Complementarity with Astrophysical Probes*

51. 08/23-30 25: **Plenary speaker**, XIX International Conference on Topics in Astroparticle and Underground Physics (TAUP2025), Xichang (CHN), *Axion cosmology*

50. 06/30-07/4 25 Axions in Stockholm 2025, Stockholm (Sweden), *indirect detection of the QCD Axion*

49. 05/26-30 25 Planck 2025, Padova (Italy), *Ultralight Black Holes as Astrophysical Particle Accelerators*

48. 03/10 25 FLASH meeting, Frascati (Italy), *WP1-Theory outputs*

47. 11/26 24 FLASH Kick-off meeting, Frascati (Italy), *FLASH kick-off meeting: Theory challenges*

46. 11/11-15 24 The 8th Shanghai Symposium on Particle Physics and Cosmology, Tsung-Dao Lee Institute (China), *Ultralight Black Holes as Astrophysical Particle Accelerators*

45. 10/15-18 24 Transient Phenomena and Physical Processes Around Supermassive Black Holes, Tsung-Dao Lee Institute (China), *New physics out of the shadow*

44. 09/03-06 24 **Plenary speaker**, 2nd General Meeting of the COST Action "Cosmic WISPers", Istanbul (Turkey), *Direct and indirect detection of the QCD Axion*

43. 07/08-12 24 15th International Workshop on the Identification of Dark Matter (IDM 2024), GSSI L'Aquila (Italy), *Indirect Detection of the QCD Axion: Resonant Conversion in Neutron Star Magnetospheres*

42. 07/01-05 24 Invisibles Workshop 2024, Bologna (Italy), H_0 *and cosmological tensions: An overview*

41. 05/15 24 FLASH Technical Design Report (TDR), Frascati (Italy), *Forecast reach of the FLASH cavity experiment at LNF*

40. 11/16-18 23 The 9th China LHC Physics Workshop (CLHCP2023), Shanghai (CHN), *Theoretical Motivations for Hidden Light Bosons*

39. 11/10-13 23 International Symposium on Cosmology and Particle Astrophysics CosPA 2023, Hong Kong, *Direct detection experiments and light bosons*

38. 10/30-11/3 23 IBS Conference on Dark World 2023, IBS Science and Culture Center, Daejeon (Korea), Direct detection experiments and light bosons

37. 09/15 23 Dark Energy from Fundamental Theories to Observations, Frascati (Italy), Direct detection experiments and light chameleons

36. 09/11-14 23 **Plenary speaker** on axions at TeVPA 2023, Napoli (Italy), The QCD axion: theory, phenomenology, and searches

35. 06/19-21 23 New Horizons in Primordial Black Hole physics (NEHOP), Napoli (Italy), *Primordial black holes and particle dark matter are intimately intertwined*

34. 06/5-9 23 Axion Across Boundaries, Galileo Galilei Institute, Arcetri (Italy), *Axions and inflation* [YouTube]

33. 02/23-24 23 Kick-off Meeting of COST Action COSMIC WISPers, INFN Frascati (Italy), *Axions and PBHs*

32. 11/18-20 22 Particle Physics and Cosmology Symposium 2022, Shanghai (CHN), *Novel Cosmological Bounds on Thermally-produced Axion-like Particles*

31. 11/29-12/03 21 **Plenary speaker**, Sixth Colombian Meeting on High Energy Physics, Santa Marta (CO), Future probes of light bosons

30. 10/12-13/21 Cosmology Frontier in Particle Physics: Astroparticle Physics and Early Universe, NCTS NTU, *Axion miniclusters in the Milky Way*

29. 07/5-16/21 Cosmology from Home 2021, *Direct detection of dark energy* [LINK to YouTube video]

28. 05/15/21 TDLI International Workshop "Current Topics on Axion", Shanghai Jiao Tong University, Shanghai (CHN), Future probes of the axion landscape

27. 10/13-16/20 The 5th IBS-IFT-MultiDark Workshop, Institute for Basic Science (IBS), Daejeon (KR), *Recent and future developments of dark matter axion physics*

26. 09/24/20 Cosmology in Colombia (CoCo 2020), Bogotá (CO), *Constraints on Reheating to SM Particles due to Large Effective Higgs Boson Mass*

25. 08/24/20 Cosmology from Home 2020, *Axion Miniclusters: Tidal Disruption and Radioastronomy* [LINK]

24. 07/21/20 13th International Workshop on the Identification of Dark Matter (IDM 2020), Zurich (CH), *The future of Axion Physics*

23. 12/2-6/19 TeVPA 2019, Sydney (AU), *Probing the Early Universe with Axion Physics*

22. 06/24-06/26 Vera Rubin Fest, Washington DC (USA), *Axions*

21. 06/10-14/19 Invisibles19, Valencia (ES), *Coordinating one of the panel discussion session*

20. 12/18/18 SLAP 2018, King's College London (UK), *The Cold Dark Matter axion and Axion Stars*

19. 12/11/18 The quest for New Physics, Instituto de Física Corpuscular (ES), *Axion Stars*

18. 09/05/18 Invisibles18 Workshop, Karlsruhe Institute of Technology (DE), *Dilute and dense axion stars*

17. 06/20/18 14th Patras Workshop, DESY Hamburg (DE),
The Higgs Boson can delay Reheating after Inflation

16. 06/12/18 Preparing for Dark Matter Particle Discovery, Chalmers University of Technology, Goteborg (SE),
The Higgs Boson can delay Reheating after Inflation

15. 03/07/18 Ultralight Dark Matter and Axions, University of Michigan (USA),
The parameter space of axion-like particles

14. 02/22/18 UCLA Dark Matter 2018, UCLA (USA),
Axions in cosmology and astrophysics

13. 08/31/17 DaνCO, CP^3 Origin (DK),
Axions and ALPs as the Cold Dark Matter

12. 08/04/17 Self-interacting dark matter, Niels Bohr Institute (DK),
Sharpening Fuzzy Dark Matter

11. 07/19/17 Advances in Theoretical Cosmology in Light of Data, Nordita (SE),
    Axion dark matter.

10. 03/5-7/17 Ultralight Dark Matter and Axions, University of Michigan (USA),
    The parameter space of axion-like particles.

9. 12/06/16 Axion Dark Matter workshop, Nordita (SE),
   Axion dark matter, miniclusters, and axion stars

8. 08/12/16 11th International Workshop on the Identification of Dark Matter (IDM 2016), Sheffield (UK),
   Galactic Cold Dark Matter from First Principles

7. 09/07/15 14th International Conference on Topics in Astroparticle and Underground Physics (TAUP 2015), Torino (IT)
   Analytical expressions for the kinetic decoupling of WIMPs

6. 11/19/14 GEOCARBON Final Meeting, Paris (FR),
   Summary contribution to GEOCARBON from Land and Ocean Components

5. 07/08/14 NEMO Users Meeting, Grenoble (FR),
   Assimilation of Physical and Carbonate Data on the Global Ocean Carbonate System

4. 10/15/10 American Physical Society Four Corners Meeting, Ogden (USA),
   An integral equation for distorted-wave amplitudes

3. 03/28/10 SnowPac & SnowCluster 2010, Alta (USA),
   Axion dark matter in non-standard cosmologies

2. 07/02/09 11th International Conference on Topics in Astroparticle and Underground Physics (TAUP 2009), Rome (IT),
   Axion cold dark matter revisited.

1. 10/20/07 American Physical Society Four Corners, Flagstaff (USA),
   Oscillation amplitude for neutrino wave packets.

Poster presentation

1. 07/13/10 The 4th international Workshop on the Interconnection between Particle Physics and Cosmology (PPC 2010), Torino (IT),
   An integral equation for distorted-wave amplitudes

Workshop attendance

November 2026	Quantum Foundations: from the Pauli Exclusion Principle to Quantum Gravity, INFN Laboratori Nazionali di Frascati (IT).
December 2025	String & Cosmology Meeting, LAPTh, Annecy (FR)
June 2025	From Puzzles to New Insights in Fundamental Physics, Campagna (IT)
May 2025	CosmoVerse@Naples, Scuola Superiore Meridionale, Napoli (IT)
May 2023	Axions across boundaries between Particle Physics, Astrophysics, Cosmology and forefront Detection Technologies, Galileo Galilei Institute, Arcetri (IT)
April 2017	The 5th MCTP Spring Symposium, Ann Arbor, MI (USA)
September 2014	Data Assimilation in Ocean Physics, Trieste (IT)
June 2013	Advanced School on Data Assimilation, Bologna (IT)
December 2009	Focus week on indirect dark matter search, IPMU Tokyo (JP)

Commissione di esame, University of Salerno

18/09/26	Laboratorio di Fisica II per Matematica (orale)
15/07/26	Laboratorio di Fisica II per Matematica (orale)
29/06/26	Cosmology & Astroparticles
23/06/26	Laboratorio di Fisica II per Matematica (orale)
19/02/26	Laboratorio di Fisica II per Matematica (orale)
11/02/26	Laboratorio di Fisica I per Chimica (scritto)
26/01/26	Laboratorio di Fisica II per Matematica (orale)

Visiting Periods at Research facilities

Aug 20–31 2026	University of Science and Technology of China, Hefei (CHN)
Jan 10–25 2026	Tsung-Dao Lee Institute, Shanghai (CHN)
Nov 19–21 2025	INFN Legnaro (IT)
Jan 7–31 2025	INFN Ferrara (IT)
Jun 09–12 2024	INFN Frascati (IT)
Feb 2–7 2024	University of Texas at Austin (TX)
Jan 17–23 2024	INFN Ferrara (IT)
Aug 21–31 2023	University of Texas at Austin (TX)
Jul 09-15 2023	INFN Frascati (IT)
Apr 20–30 2019	University of Michigan (MI)
Apr 10–20 2019	Barry University (FL)
Jan 20–31 2019	INFN Frascati (IT)
May 1–15 2017	Perimeter Institute (CA)
Feb 25–28 2017	Massachusetts Institute of Technology (MA)
Jan–Mar 2017	University of Michigan (MI)

Visitors hosted at the University of Salerno

9-10 Nov 2026	Dr. Mariia Khelashvili, SISSA (IT).
19-21 Oct 2026	Dr. Michael Zantedeschi, INFN Pisa (IT). FARB 2023 Prof. Blasone
15-19 Dec 2025	Ms. Agnese Tolino, IFIC, Valencia (ES). INFN grant from Prof. Lambiase
15-19 Dec 2025	Dr. Valentina De Romeri, IFIC, Valencia (ES). INFN grant from Prof. Lambiase
12-31 Oct 2025	Ms. Kratika Mazde, AIP, Paris (FR). INFN grant from Prof. Lambiase
9-16 June 2025	Dr. Hong-Yi Zhang, Tsung-Dao Lee Inst., Shanghai (CHN)
20-24 May 2025	Mr. Hanyu Cheng, Tsung-Dao Lee Inst., Shanghai (CHN)
27-30 April 2025	Prof. Momchil Naydenov, Sofia University (BG)
10-16 April 2025	Dr. Yifan Chen, Niels Bohr Institutet, Copenhagen (DK)

Visitors hosted at the Tsung-Dao Lee Institute

Jan 2 2025	Prof. Giorgio Torrieri, University of Campinas (BR)
Dec 23-25 2024	Dr. Jiajun Chen, Chongqing Southwest University (CHN)
Dec 16-19 2024	Dr. Tucker Manton, Hangzhou University (CHN)
Dec 9 2024	Prof. Salvatore Capozziello, University of Napoli (IT)
Dec 9 2024	Dr. Qingyu Gan, University of Napoli (IT)
Nov 14–19 2024	Prof. Stefano Gariazzo, University of Torino (IT)
Oct 27–31 2024	Dr. Michele Tammaro, INFN Firenze (IT)
Apr-May 2024	Prof. Enrico Schiappacasse, San Sebastian University (CL)
Mar-May 2024	Andrea Boccia, University of Napoli & Scuola Superiore Meridionale (IT)
Apr 12 2024	Prof. Jessica Turner, University of Edinburgh (UK)
Feb 25–27 2024	Prof. Alejandro Ibarra, Technical University of Münich (DE)
Feb 19–23 2024	Dr. Eleonora Di Valentino, University of Sheffield (UK)
Nov 7–11 2023	Dr. Florian Kühnel, Ludwig-Maximilians Universität (DE)
Sep 16–28 2023	Prof. Fabio Iocco, University of Napoli (IT)

Books

June 2024 Co-author, with Enzo Pellegrino, of Dove Abita il Tempo (Edizioni Pendragon, 2024), an Italian-language book connecting physics and philosophy for a non-specialist audience. Link to the book's website with Edizioni Pendragon

Book presentation

Feb 13 2025 Presentazione "Dove Abita il Tempo", Gallery 16, Bologna (Italy). Link

Jul 29 2024 Presentazione "Dove Abita il Tempo", Palazzo del Trono, Cetraro, CS (Italy). Link

May 31 2024 Presentazione "Dove Abita il Tempo", Sala Primavera, Loiano, BO (Italy). Link

Terza Missione (Outreach)

Oct 2026 Invited speaker, Space Week Seminar 2026.

Oct 2026 Participant, European Night of Researchers, Villa Comunale, Salerno (Italy).

Jan 2026 Lecture: *"Il cielo stellato sopra di noi: viaggio tra filosofia e astrofisica"*, Circolo San Carlo, Bologna (Italy).

Oct 2025 Participant, European Night of Researchers, Villa Comunale, Salerno (Italy).

Jan 22 2025 Lecture: *"Quando muore una stella"*, Circolo San Carlo, Bologna (Italy). Link of the event

Jul 30 2024 Lecture, Giardino Palazzo del Trono, Cetraro, CS (Italy). Link of the event

Jul 2022 South physics observatory astronomy summer camp "Astronomy for Future Scientists", The University of Utah. Host: Paul Ricketts.

Jan 2022 Virtual interview with the online newspaper "Fanpage" on dark energy (in Italian) [Youtube link]

September 2021 Interview for the podcast "co.scienza" (in Italian), aimed at young researchers [Link]

August 2021 Public lecture "Le cinque ere dell'Universo e la vita su altri pianeti" (In Italian). Youtube links: PART 1 PART 2

August 2020 Public lecture "Buchi neri e onde gravitazionali: La ricerca in Italia" (In Italian). Youtube links: PART 1 PART 2

March 2016 Public lecture "Capire le onde gravitazionali" (In Italian), Bologna.

November 2015 Public lecture "Physics for everybody!" (In Italian), Bologna.

February 2013 Public lecture "Galileo, father of the scientific method" (In Italian), Bologna.

April 2011 Public lecture "Dark Matters", The University of Utah, Math Department.

Press coverage and media appearance

August 2025 "Neutrino masses are not likely to originate from interactions with dark matter", **Phys.org**

July 2025 "Can Dark Matter Explain Neutrino Mass? TDLI Team Presents Contradictory Evidence", **Shanghai Jiao Tong University press release**

October 2024 "Sulle tracce della quinta forza", **University of Trento press release**

September 2024 "Exploring a Fifth Force via the Trajectory of Asteroid Bennu", **Shanghai Jiao Tong University press release**

October 2021 "Signal From The XENON1T Experiment May Be A Hallmark Of Dark Energy", **Forbes**

September 2021 "Have we detected dark energy?", **www.cam.ac.uk**

June 2021 "Can we see dark energy from Earth? New experiments offer hope", **Space.com**

January 2021 "Black holes could reach 'stupendously large' sizes", **Queen Mary University of London**

November 2020 "Top ArXiv papers from week 48 2020, entry #2", **Sunny Vagnozzi's blog**

November 2020 "Top ArXiv papers from week 46 2020, entry #2", **Sunny Vagnozzi's blog**

September 2020 "'Stupendously large' black holes could grow to truly monstrous sizes", **Space.com**

September 2020 "Black holes - Do they grow stupendously large?", **Astrobites.org**

April 2020 "Interview with Luca Visinelli on "Il Messaggero" (In Italian)", **ilmessaggero.it**

September 2018 "Hunting for extra dimensions with gravitational waves", **Oskar Klein Centre blog**

August 2018 "Zoos, Swamplands and Cosmology", **Astrobites.org**

September 2017 "Interview with Luca Visinelli", **Oskar Klein Centre blog**

Additional work experience

2015 **Editing work for JCAP and JHEP**, Sissa-Medialab publications, Trieste (Italy)

2015 – 2016 **High school teacher**, Public High School "E. Majorana", Bologna (Italy)
The work contract is available [at this link].

2015 – 2016 **High school teacher**, Private High School "M. Malpighi", Bologna (Italy)

2011 – 2016 **Private tutoring** in mathematics and physics for high school and university students.

2012 – 2013 **Quantitative Analyst**, Iason LTD, Milan www.iasonltd.com.
I have implemented a set of numerical codes aimed at pricing defaultable coupon bonds in C/C++, Matlab, Excel. My personal contribution has been cited in the book:
A. Castagna and F. Fede, *Measuring and Managing Liquidity Risk*, Wiley (2013).
The work contract is available [at this link].

2011 – 2012 **Editor for high-school mathbooks**, Zanichelli Editore. Textbook here.

Other Experiences

2015 – present **Voluntary affiliation, Italian Civil Protection Department**, *Loiano (Bologna)*, Italy.

2014 – 2016 **Elected as Town Councilor**, *Comune di Loiano (Bologna)*, Italy.

Professional references addressing my research activity

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Professional references addressing my teaching activity

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Additional references

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Email: tsutomu.tyanagida@sjtu.edu.cn.

Frank Wilczek
Full Professor,
• Department of Physics, Massachusetts Institute of Technology
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• Department of Physics and Origins Project, Arizona State University
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Autorizzo il trattamento dei miei dati personali presenti nel curriculum vitae ai sensi del Decreto Legislativo 30 giugno 2003, n. 196 e del GDPR (Regolamento UE 2016/679).

October 27, 2026 Luca Visinelli